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Accelerating Well Reactivation Through Digital Transformation: Advancing Operational Visibility and Decision-Making with Smart Inactive Well Management Systems in Upstream Operations

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Abstract

This paper outlines the development and implementation of a comprehensive, standardized, and data driven well management system designed to optimize the end to end well reactivation process. The scope encompasses the integration of advanced digital technologies, including data analytics, automated workflows, and smart systems, to enhance decision making, improve operational efficiency, and reduce reactivation cycle time across multiple operational Assets through the establishment of a unified framework that enables consistent and objective review and evaluation of candidate wells.

The process began with the implementation of a unified well problems categorization methodology, governed by automated validation workflows, multi-source data and systems integration including historical performance data, subsurface attributes, production trends and Well integrity parameters to systematically evaluate candidate wells for reactivation. Each step was designed for minimal manual intervention, supported by built-in logic, exception handling aligned with governance standards. Throughout the process, advanced analytics and visualization tools were integrated to deliver actionable insights to multidisciplinary teams. Dashboards provided visibility into KPIs, reactivation potential, and bottlenecks, enabling proactive and collaborative decision making from cross disciplinary teams.

This methodology established a closed loop system that facilitated a 20% acceleration in well reactivation cycles since its adoption in 2020, while delivering an average gain of 25 MBD from the reactivation of inactive strings during 2024. Additionally, a 30% reduction in the number of inactive strings was achieved last year, resulting in historically low inventory levels, with less than 2% of wells remaining inactive. This reduction was a result of rigorous integrity management practices and continuous monitoring of the well conditions. This framework has been designed to adapt to various operational scenarios and can be easily implemented across different Assets/ fields, ensuring consistency and reliability in well reactivation efforts. In conclusion, this initiative validated that data-driven standardization and automation are key enablers of efficient, scalable, and auditable management of inactive well portfolios across upstream operations. The

success of this initiative underscores the importance of leveraging data and technology to drive operational excellence and achieve sustainable results in the oil and gas industry.

This paper introduces a novel, integrative framework on well reactivation focused on a holistic, end to end solution that unifies key components of the well reactivation process into a standardized, smart, and scalable well management system. The key contribution lies in the standardization of reactivation processes across diverse operational Assets, enabled by the creation of a unified screening methodology and automated decision support tools. Unlike isolated digital solutions, this approach demonstrates how automated data integration and workflow orchestration can be institutionalized across organizations, reducing subjectivity, improving consistency, and enhancing operational agility. The outstanding results further validate the system's impact, offering a replicable model that can be adapted to other organizations seeking to modernize and streamline their well reactivation programs.

Introduction

The effective management of inactive wells remains a critical challenge for upstream oil and gas operations, particularly in mature fields where production optimization and asset longevity are strategic imperatives. The management of inactive wells is a complex and critical domain. The causes of inactivity can be related to the reservoir, well, or facility, and can pose major integrity and HSE risks. Reactivating them requires cross-functional collaboration, dedicated resources, advanced technologies, and skilled personnel.

This paper presents a comprehensive, data-driven well management system and its latest developments designed as an integrated ecosystem to streamline the analysis and reactivation of inactive wells with the key stakeholders, processes and standardized practices as enablers to accelerate well reactivations. By centralizing data, methodologies, and cross-disciplinary collaboration, this system enables the seamless alignment of technical workflows, unifying reservoir, production, and operations teams under a standardized and responsive framework.

Through the application of advanced analytics, structured decision-making processes, and real-time performance tracking, the system fosters a proactive environment for identifying reactivation opportunities, prioritizing interventions, and accelerating the implementation of field actions. The resulting ecosystem not only enhances operational efficiency and well performance recovery but also reinforces organizational agility by bridging data silos and promoting consistent, collaborative practices across all involved disciplines across the company's assets.

The latest development posed to transform our business operations is the Inactive Strings Management System (ISM). This innovative, web based system has addressed numerous challenges related to Operational Excellence and has marked a significant impact to achieve the Inactive strings reactivation targets for the company. With the full deployment and continuous enhancement of the ISM, we are witnessing a major shift in how we handle data integrity issues, lack of automation, and system integration. It has also improved our tracking of well reactivation actions, providing a systematic approach to prompt multidisciplinary well revisions. The standardized platform for managing inactive strings ensures that the best practices and lessons learned are shared across all company assets.

The increased visibility of well problems and the enhanced accountability of key stakeholders have been remarkable. ISM facilitates the firming, maturing, and acceleration of well reactivation actions in an automated and collaborative environment. This not only reduces silos but also ensures alignment, knowledge sharing, and effective information dissemination.

Moreover, the improved resource planning and production gains achieved with ISM come without compromising the integrity of wells and facilities. This level of innovation and collaboration marks a significant step forward for our company.

Inactive Strings Management Process

The Management Process starts with following closely the well activities on daily bases once the well is not capable to flow for 7 days (168 hours) in a calendar month is considered an Inactive string, then the well shall be assigned to one of the Three (3) main "Inactive" categories (well, facilities or reservoir related issues) and an action category as below:

- Investigation
- Rig Work-over
- Rigless / Barge Work
- Surface Work
- Tie-in

When any well did not flow for more than 7 days (168 hours) in a calendar month irrespective of the reason; whether it is incapable of flowing naturally (If not under one of the Artificial Lift schemes) or intentionally shut-down, where a decision was taken to shut-in the well for whatever reason and considering the well remains inactive at the end of the current month, it needs to be captured in the system and proper inactive string category assigned.

Well performance is monitored to ensure it is operating within the desired production / injection envelop in a healthy manner. When symptoms of abnormalities appear, then the well is classified as "active problematic" for closely monitoring and investigation. If the well becomes inactive, investigation is continued, and main problem, inactive category, detailed reactivation plan, action type are registered and tracked. Well reactivation is prioritized based on integrity Status, production / injection gain, and economics indicators.

Figure 1 shows that key stakeholders from divisions such as Reservoir, Production Optimization, Well Integrity, Operations, Drilling, and Project & Construction play vital roles in the system. Effective collaboration and communication are essential for successful reactivation, meeting production targets, and maintaining HSE compliance.

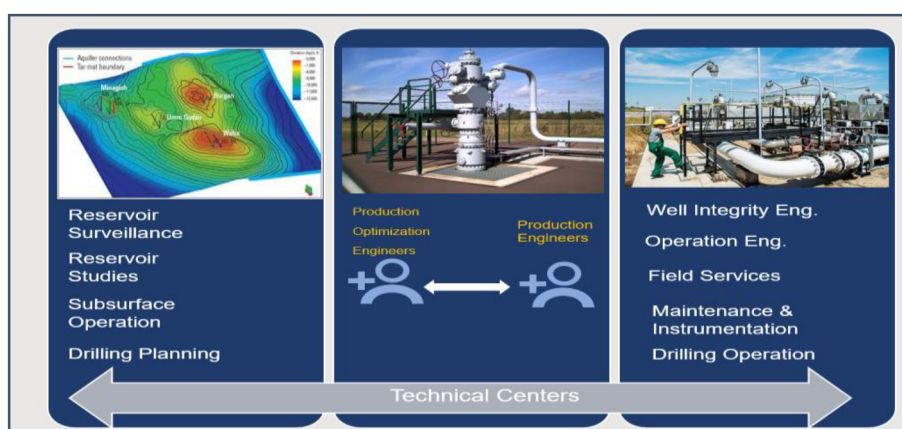


Figure 1—Inactive Strings stakeholders

Inactive Strings Management System (ISM)

This integrated centralized solution, deployed across all company assets, streamlines the well reactivation cycle by monitoring phases such as Inactive Register, Mitigation, Action Tracking, and Execution. Featuring web-based tools and integration with risk assessment, drilling schedules, and location maps systems, it

standardizes inactive string management, aligns with ADNOC guidelines, and helps meet business KPI targets.

Comprehensive Modular Design

The system includes dedicated modules covering the entire well reactivation cycle, such as data entry, validation, live inventory, tracking, dashboards, prioritization, and Performance KPI monitoring. ISM includes automated workflows and integration with core systems, adhering to guidelines. ISM generates automatic notifications for pending data entries, validations, approvals, and overdue activations. A sample of the ISM Navigation page is shown in [Figure 2](#).

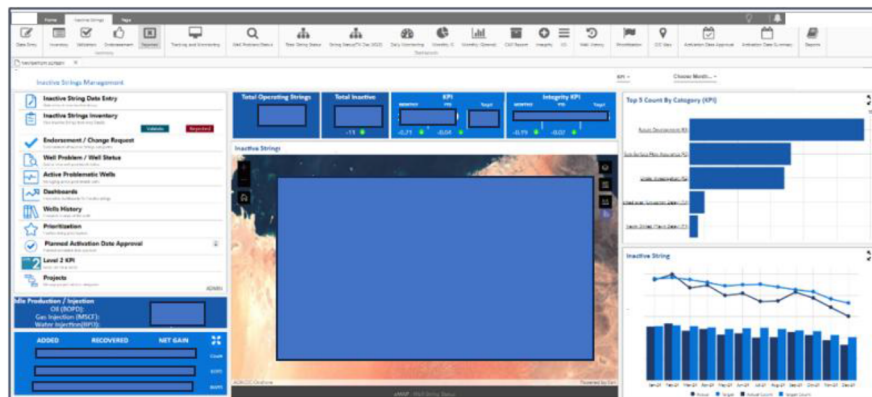


Figure 2—Inactive Strings Management System

This system is considered as the backbone and source true for Inactive Strings Management within ADNOC Onshore. It has a high level of data integrity and reliability. It has led to major improvements on the current business and system processes within ADNOC Onshore by having a common, unique platform to manage the inactive strings through a standardization of different practices across the Assets and incorporation of best practices and lessons learned, led by proper multidisciplinary teams of subject matter experts.

Enhanced business process

A Holistic approach was adopted by enhancing the existing business process and system to manage de Inactive strings. A comprehensive review of the existing processes was conducted with the involvement of both internal and external stakeholders. The major findings indicated that the wells register requires further validation and assurance processes, as well as approval workflows to accurately confirm reasons for Inactivity and activation plans with the appropriate responsible authorities. A multidisciplinary task force was established within the company, comprising subject matter experts from all functions. Eight dedicated subcommittees were formed to cover the main inactive categories groupings, leading the detailed review, validation, and monitoring of each inactive string's reactivation actions with the applicable assigned related subject matter experts including Technical Centers. Each category custodian to be responsible to always assure the accuracy of the Well register with the accountability on the Asset's well custodians.

An strategic framework was developed aiming to enhance the monitoring of the wells, improving wells and actions visibility to all levels aiming to reduce the Inactive strings stock and unlock potential by addressing the objectives below:

- Unified well problems categorization methodology.
- Dedicated verification by each subcommittee by assigned Leader & verifier (Allocated SME's) for each category.

- Implementation of New Validation workflows for core Inactive strings categories.
- Live process for Inactive strings Register, Validation and Reactivation tracking.
- Proactive identification of Potential Inactive strings (Active Problematic Wells)
- Monitoring of Activation date compliance as per Inactive strings guidelines requirements.
- Closer monitoring of Newly drilled wells part of Projects.

Enhancements to the Inactive Strings Management System (ISM)

To address the identified challenges and achieve the ambitious targets for reducing inactive strings, the ISM system underwent significant enhancements. These improvements were designed to streamline processes and serve as a critical enabler of the strategic framework. Some of the notable changes are described in the following sections including advanced analytics and visualization tools providing actionable insights to teams, improving decision making and collaboration. Moreover, this enhancement promoted proactive and collaborative decision-making among multidisciplinary teams through enhanced visibility into key metrics.

Implementing New validation workflows

A new automated workflow was implemented to bring to the validation of technical authorities and subject matter experts (SME's) as part of the approval process for core categories including Well integrity, newly drilled wells part of Projects and Reservoir related. The workflow below describes the process from initiation by the corresponding Asset focal point to the SME's review and validations. All changes are reflected in real time on the Inactive strings inventory and well register.

These enhancements reduced subsurface well integrity issues and minimized their impact on the Inactive Strings KPI by enabling prompt technical assessments, allowing risk downgrades by killing and securing the wells reducing annulus pressure or continued operation where appropriate, especially for wells with sustained annulus pressure. They also facilitated ongoing risk assessments for all inactive strings, increased visibility of wells awaiting tie-in and commissioning, streamlined asset development, drilling, and engineering activities, and automated workflow notifications according to approval sequences. The Overall process and key contributors of this workflow is shown in Figure 3 below.

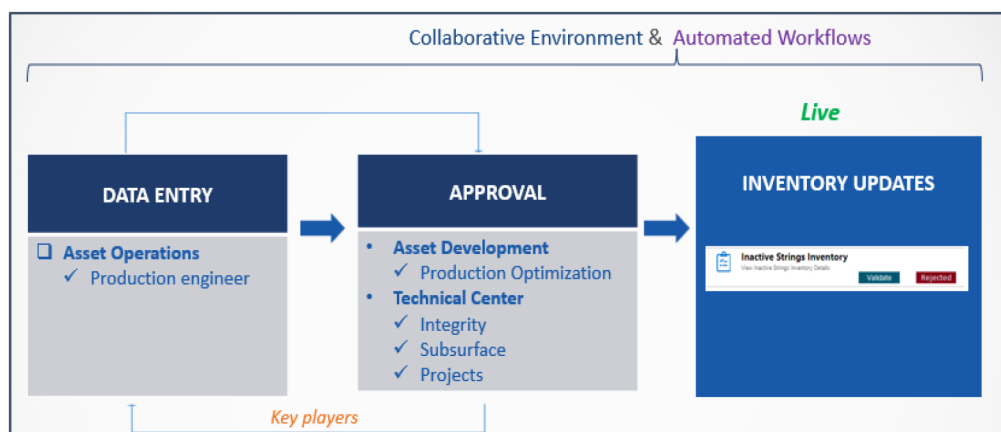


Figure 3—New categorization approval workflow

Activation compliance, Noncompliance wells

Inactive strings, when not managed effectively, can lead to significant delays in well reactivation, impacting resource allocation and budget planning. The Inactive strings guidelines stipulate a maximum allowable duration for each inactive category, which, if exceeded, necessitates immediate corrective actions. However,

a substantial number of strings were found to be exceeding this duration without proper identification, monitoring, and timely intervention.

Automated workflows, dashboards and notifications were implemented into the ISM system increasing the visibility of these cases to be monitored to expediate the reactivations. Monthly reviews of overdue plan activations are conducted by Asset Managers, who are required to provide justifications for any delays and update the reactivation plans accordingly. This process is now monitored by the ISM system, which captures and tracks all modifications and justifications, ensuring that necessary actions are expedited. A Monitoring page for Non Compliance wells is shown in Figure 4 below.

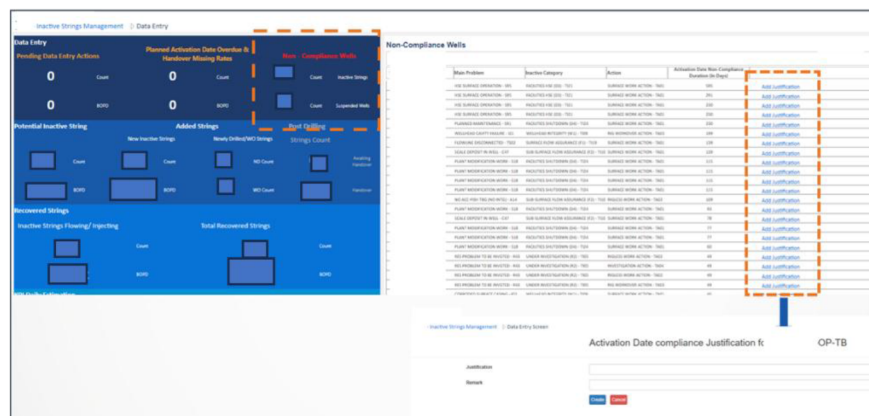


Figure 4—Activation compliance monitoring workflows

Live Inventory Tracking

Live inventory tracking of inactive strings enhances visibility, monitoring, resource allocation, decision-making, and compliance, leading to more efficient and timely well reactivations. The Net Inventories can be positive or negative at the end of each Month depending on the number of strings becoming Inactive and the number of strings reactivated during that period.

Visibility and tracking provide up-to-date information on the status of inactive strings, allowing for immediate identification of any deviations from the prescribed durations. This ensures that any issues can be addressed promptly, reducing the risk of delays in reactivation. Daily monitoring of Inactive strings allows for more informed decision-making. Asset managers can quickly assess the status of inactive strings stocks and make decisions based on the most current information, leading to more effective management of well assets.

By having a live inventory, asset managers can better allocate resources and budget for reactivation activities. This optimized allocation helps in avoiding unnecessary expenditures and ensures that resources are directed towards the most critical areas. With Live dashboards and notification mechanisms, stakeholders can continuously monitor the status of inactive strings. This continuous monitoring helps in maintaining compliance with Inactive Strings Management (ISM) guidelines and ensures that corrective actions are taken in a timely manner.

Moreover, by continuously tracking and re-categorizing inactive strings, the system ensures that the status of each string is accurately reflected. This reduces the risk of misclassification and ensures that appropriate actions are taken based on actual status. A sample of Live inventory tracking dashboard is shown in Figure 5 below.

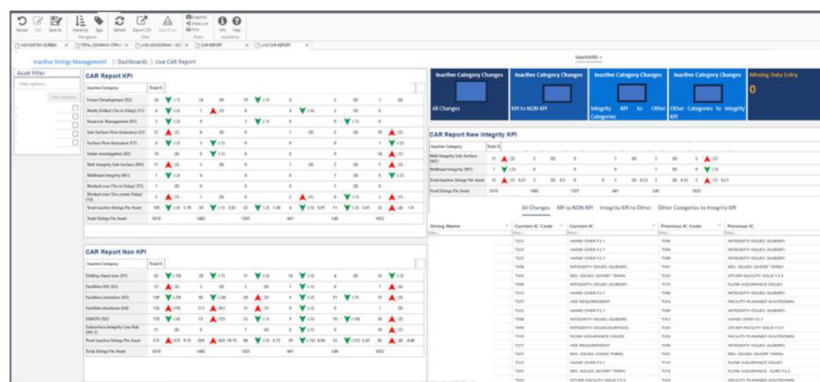


Figure 5—Live Inventory tracking dashboard

New Wells Awaiting Handover

The transition or handover of well assets from drilling to operations or engineering teams is a critical phase in the lifecycle of well development. The implementation of automated workflows within the Inactive Strings Management system to enhance the monitoring and tracking of wells post-rig release, thereby expediting the handover process and mitigating potential delays.

Upon the completion of new well or workover drilling and completion activities, and subsequent rig release, a handover process from drilling to operations or engineering teams is initiated. This process is essential for completing tie-in and commissioning activities. However, challenges arise when wells are drilled as part of development plans while production facilities are still under construction, undergoing upgrades, or not yet commissioned as part of major engineering projects.

A significant issue identified was the lag in the handover process post-rig release, with wells not being flagged in the system for necessary actions. This lack of tracking and visibility resulted in delays, leading to substantial locked-in potential.

To address these challenges, automated workflows were embedded within the ISM system. These workflows systematically monitor and track wells from rig release, automatically assigning them to an "Awaiting Handover" category. Dedicated dashboards and automated notifications were implemented to alert assets and engineering focal points, facilitating close tracking and expediting tie-in and commissioning activities. The implementation of automated workflows within the ISM system has significantly improved the visibility and management of wells post-rig release. The systematic tracking and categorization of wells have ensured timely interventions, reducing the risk of delays. Additionally, wells facing integrity issues were promptly identified, and actions were taken to minimize risks and secure these wells.

Monthly reviews of overdue plan activations are conducted by asset managers, who provide justifications for any delays and update the reactivation plans accordingly. This process is monitored by the ISM system, capturing and tracking all modifications and justifications, ensuring that necessary actions are expedited leading to more efficient and timely handovers. This digital transformation has not only improved compliance with ISM guidelines but also optimized resource and budget allocation, ultimately contributing to the overall efficiency of well operations.

Moreover, this new workflow was integrated with a dedicated Projects Module to track the progress of the engineering activities and readiness of the production facilities once the wells are completed handover to engineering.

The duration of the awaiting handover period was significantly reduced to a remarkable 50% from an average of 165 days to 75 days. The acceleration of handovers was crucial in unlocking significant production potential by accelerating the delivery of strings during 2024 as part of Project wells tie-in and engineering activities. Below [figure 6](#) exhibits an snapshot of Project New drilled wells tracking Dashboard in the ISM system.

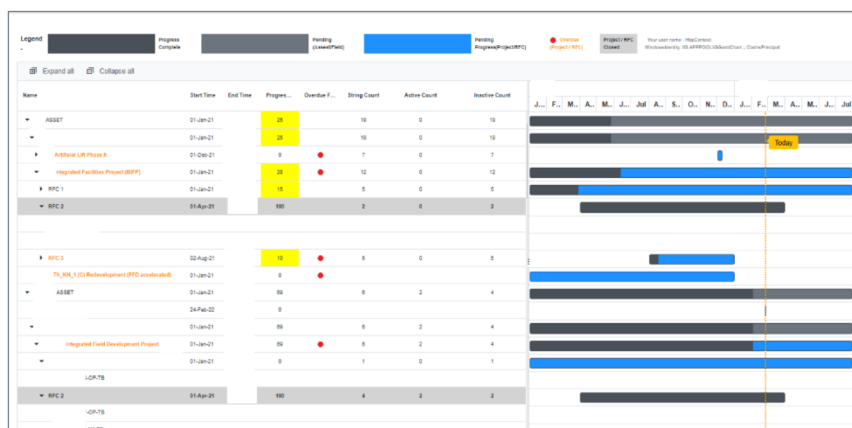


Figure 6—Project New drilled wells tracking Dashboard

Identify Potential Inactive strings (Active problematic wells)

The proactive and smart detection of problematic wells promoted to expedite well investigations that led to reduction of the Inactive strings stock. This dedicated module was enhanced to identify Active problematic wells, which considers flowing wells that exhibits one or more of the following issues that can significantly impact the performance and integrity of the well, necessitating timely interventions and corrective actions to mitigate potential risks and optimise well operations. Each step was designed for minimal manual intervention, supported by built-in logic, except for handling:

- **Well & Facilities Integrity Problems:** This includes any structural or operational issues that compromise the integrity of the well or associated facilities.
- **Significant Abnormal Well Head Parameters:** Any deviations in the well head parameters that are outside the normal operating range.
- **Fluctuating Flow / Instability:** Irregularities in the flow rate or stability of the well, which can impact production.
- **Well Testing Data Abnormalities:** Any anomalies or inconsistencies observed in the data collected during well testing.
- **Loss of Well Potential:** A reduction in the well's ability to produce at its expected capacity.
- **Inability to Meet Production / Injection Targets:** Failure to achieve the set production or injection targets for the well.

Proactive approach. The system maintains a continues monitoring of Active wells with signs of problems, prompting early well investigations and mitigation actions. Over 300 strings during 2024 were investigated and actions were taken by Assets to minimize the Inactive strings inventory. Figure 7 shows a sample of Potential problematic wells analyzer screens in ISM system.



Figure 7—Potential problematic wells analyzer

Achievements and Business Benefits

The continuous evolution of the Inactive Strings Management processes and system in ADNOC Onshore has driven these positive results and yielded significant benefits to the company:

Enhanced Performance. The implementation of an enhanced Inactive strings Management process and governance has resulted in an average 30% reduction in Inactive strings stock and a more remarkable 50% reduction for Inactive strings with Integrity issues during 2024. This proactive monitoring of well conditions has minimized historical inventory levels to less than 2% of inactive wells stock as can be seen on Figure 8 below.

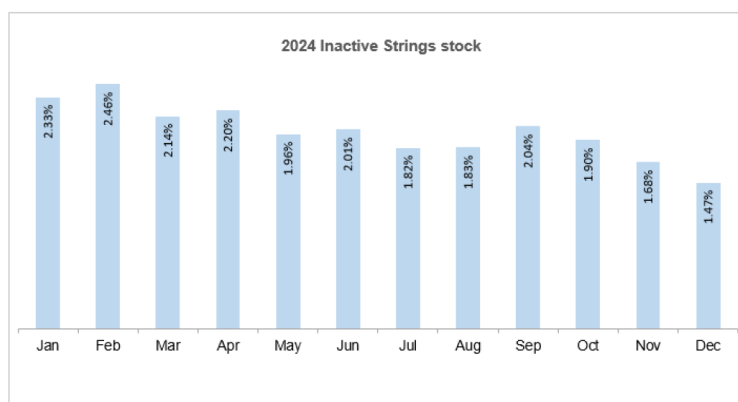


Figure 8—2024 Inactive strings stock.

Accelerated Reactivation Cycles:

The system has supported the acceleration of the well reactivation cycle by 20% since its implementation in 2020, leading to 25 MBD average gain from inactive strings reactivation during 2024. The duration of the awaiting handover period was significantly reduced 50% from an average of 160 days to 80 days. The acceleration of handovers was crucial to increase the visibility of these wells to the parties concerned to accelerate the further engineering, tie-in and commissioning activities, supporting the delivery of more than 100 strings during 2024 as part of Project wells tie-in and engineering activities.

Proactive approach. The system maintains a continues monitoring of Active wells with signs of problems, prompting early well investigations and mitigation actions. Over 300 strings during 2024 were investigated and actions were taken by Assets to minimize the Inactive strings inventory.

Improved Accuracy and Compliance. The system enhancements have led to improved accuracy in data collection and analysis, ensuring better compliance with corporate inactive strings guidelines, company standards and best practices across Company Assets.

Improved Accountability. It has increased problem visibility and accountability of key stakeholders to firm, mature and accelerate well reactivation actions in a collaborative environment that has contributed to reduce silos ensuring stakeholder alignment and integration, sharing knowledge and information more effectively.

Digital Transformation. This solution is part of a digital transformation strategy to revamp our production management systems. Proven benefits include transforming our customers' experience by implementing web-based functionalities, a user's friendly interface, intuitive design, increasing speed and agility data-based insights. Encourages collaboration across departments, fosters a digital culture deploying more efficient processes and systems to react to rapid business changes. The transformation of a lengthy end-to-end process makes it more effective and steadfast.

Conclusion

In conclusion, this framework has demonstrated its ease of implementation across various assets and fields, ensuring consistency and reliability in well reactivation efforts. The initiative has validated that data driven standardization and automation are key enablers of efficient, scalable, and auditable management of inactive well portfolios across upstream operations. The success of this initiative underscores the importance of leveraging data and technology to drive operational excellence and achieve sustainable results in the oil and gas industry. Moreover, it highlights that digital transformation is crucial in streamlining processes and enhancing efficiency in well management practices.

The increased visibility of well problems and the enhanced accountability of key stakeholders have been remarkable. The Integrated Solution Model (ISM) facilitates the firming, maturing, and acceleration of well reactivation actions in an automated and collaborative environment. This not only reduces silos but also ensures alignment, knowledge sharing, and effective information dissemination.

Additionally, the improved resource planning and production gains achieved with ISM have been accomplished without compromising the integrity of wells and facilities. This level of innovation and collaboration marks a significant step forward for our company. Implementing comprehensive management systems helps stakeholders achieve better performance and compliance in their operations. The enhanced collaboration among stakeholders leads to improved compliance and successful outcomes in well management.

Future enhancements will further empower the tool with advanced features such as generative AI, e-learning training, and portable devices. Hence, the solution is laying down the foundation to sustain growth and maintain a competitive edge in the digital age by applying deeper and more targeted analytics that enable better decision-making.

Acknowledgments

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Nomenclature

ADNOC	Abu Dhabi National Oil Company
ISM	Inactive Strings Management (System)
KPI	Key Performance Indicator

HSE Health, Safety, and Environment
SME Subject Matter Expert
MBD Thousand Barrels per Day
AI Artificial Intelligence

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